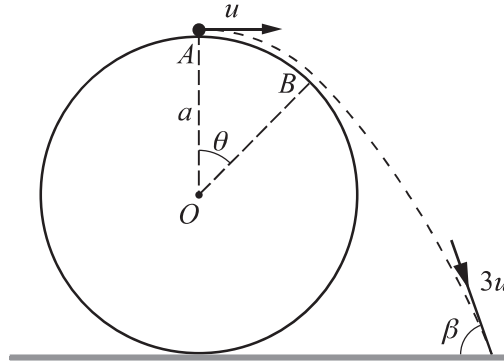




A fixed smooth sphere with radius a and centre O rests on horizontal ground. A particle is projected horizontally from the highest point, A , of the sphere with speed u . The particle begins to move in a vertical circle along the surface of the sphere. The particle loses contact with the sphere at the point B , where the angle AOB is θ .

After leaving the surface of the sphere, the particle moves freely under gravity before striking the horizontal ground with speed $3u$ at an angle β to the horizontal (see diagram).

Find the value of β . [8]

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